Deep Impact and EPOXI Missions Namespace

NASA Planetary Data System

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USER GUIDE

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The Deep Impact mission, launched 12 January 2005, encountered comet 9P/Tempel 1 in July of that same year, sending the Impactor Spacecraft into the surface of the comet on 4 July 2005 while the Flyby spacecraft observed the effects of the impact. The Deep Impact Flyby spacecraft continued on a a compbined extended mission known as "EPOXI", a portmanteau of two extended investigations: Extrasolar Planet Observation and Characterization (EPOCh); and the Deep Impact Extended Investigation (DIXI) of Comet 103P/Hartley 2 in 2010 and, in 2013, Comet C/2012 S1 (ISON). This DI dictionary was created as part of the conversion of the final archival products of the Deep Impact and EPOXI missions from PDS3 to PDS4 format.

CHAPTER

ONE

INTRODUCTION

This *User's Guide* provides a brief overview of the Deep Impact and EPOXI mission namespace, identified by "DI" or "di:", for those working with data from either the Deep Impact (DI) primary or EPOXI extended missions.

Note *that the Deep Impact/EPOXI legacy data migration is ongoing, with labels being designed for each instrument in turn. This namespace is in active development and will continue to be so for the forseeable future.*

Data from the primary and extended missions were all produced by the same pipeline in PDS3 format. They were converted to PDS4 format more than 10 years after the last PDS3 data products were archived, with some assistance from scientists who worked on the original Deep Impact and DI Extended missions.

This guide presents the major features of the namespace.

CHAPTER

TWO

OVERVIEW OF THE DEEP IMPACT AND EPOXI MISSION DICTIONARY

The Deep Impact mission, launched 12 January 2005, encountered comet 9P/Tempel 1 in July of that same year, sending the Impactor Spacecraft into the surface of the comet on 4 July 2005 while the Flyby spacecraft observed the effects of the impact. The Deep Impact Flyby spacecraft continued on a a compbined extended mission known as "EPOXI", a portmanteau of two extended investigations: Extrasolar Planet Observation and Characterization (EPOCh); and the Deep Impact Extended Investigation (DIXI) of Comet 103P/Hartley 2 in 2010 and, in 2013, Comet C/2012 S1 (ISON). This DI dictionary was created as part of the conversion of the final archival products of the Deep Impact and EPOXI missions from PDS3 to PDS4 format.

- **Primary Steward:** Anne Raugh, Small Bodies Node, University of Maryland (@acraugh on Github)
- **Dictionary Repo:** https://github.com/pds-data-dictionaries/ldd-di
- **Namespace Prefix:** di:

Corrections, changes, and additions should be submitted through the PDS LDD Issue Repo.

CHAPTER

THREE

ORGANIZATION OF CLASSES AND ATTRIBUTES

The Deep Impact/EPOXI dictionary has a single top-level class that must be used to access any of the DI metadata classes. Below that, there are major subclasses for metadata of related to various common aspects of the observations (for example, observing geometry, spacecraft environment, or observation planning). Calibrated data and data from the exoplanet observations have additional classes to provide specific details for those products.

The following sections describe the major divisions of the Deep Impact/EPOXI mission namespace, in the order in which they occur in the schema (and thus, labels).

3.1 Top-Level Class: <di:Mission_Parameters>

The *<di:Mission_Parameters>* class acts as a wrapper for all other DI classes. It contains four attributes and three subclasses.

The sequence of elements and classes within the *<di:Mission_Parameters>* class is:

- *<di:mission_phase_name>*
- *<di:spacecraft_clock_start>*
- *<di:spacecraft_clock_mid>*
- *<di:spacecraft_clock_stop>*
- *<di:Observation_Parameters>*
- *<di:Instrument_Parameters>*
- *<di:Image_Statistics>*

You can see a complete outline of the namespace under the *Deep Impact/EPOXI Mission Namespace Outline* topic.

3.1.1 Attributes of <di:Mission_Parameters>

<di:mission_phase_name>
: This attribute contains the name of the mission phase as reported by the pipeline. This is human-readable.

<di:spacecraft_clock_start>
: A string representing the onboard clock string at the start of the observations.

<di:spacecraft_clock_mid>
: A string representing the onboard clock string at the midpoint of the observations.

<di:spacecraft_clock_stop>
: A string representing the onboard clock string at the end of the observations.

3.1.2 Subclass: <di:Observation_Parameters>

The *<di:Observation_Parameters>* class provides attributes describing the intent and circumstances of the observation that produced the data in the product. As in the PDS common namespace, in the DI dictionary attribute names are all lowercase; class names are in title case.

This class contains:

- <di:observation_id>
- <di:observation_description>
- <di:mission_activity>
- <di:product_type>
- <di:spacecraft_processor_id>
- <di:sdc_pipeline_file_name>
- <di:Imaging>
- <di:Geometry>

<di:observation_id>, <di:observation_description>, and <di:mission_activity>
These attributes identify the purpose of the observation and link it to mission logs.

<di:product_type>
This attribute is used to indicate the specific processing level of the associated data product. In particular, it identifies reversibly calibrated data as distinct from cleaned data for which the processing is not reversible.

<di:spacecraft_processor_id>
This attributes which of two onboard processors obtained the data.

<di:sdc_pipeline_file_name>
This attribute is provided to help document the provenance of the data as it passed through the Science Data Center.

<di:Imaging>
This subclass contains attributes related to onboard image acquisition.

<di:Geometry>
This subclass contain attributes related to observational geometry that are not included in the Geometry discipline namespace. In particular, it includes a vector defining the rotation rate of the spacecraft.

3.1.3 Subclass: <di:Instrument_Parameters>

The *<di:Instrument_Parameters>* subclass provides attributes specific to the instrument commanding and environment during the observation.

This class contains:

- <di:instrument_mode>
- <di:instrument_mode_mnemonic>
- <di:instrument_mode_id>
- <di:Instrument_Temperatures>
- <di:Instrument_Voltages>

<di:instrument_mode>, <di:instrument_mode_mnemonic>, <di:instrument_mode_id>
These attributes provide the same instrument mode setting information in three different forms. *<di:instrument_mode>* is the mode human-readable form.

<di:Instrument_Temperatures>
This class presents a series of temperature readings from sensors in specified locations on the spacecraft.

<di:Instrument_Voltages>
This class presents a series of voltage readings from sensors in specified locations on the spacecraft.

3.1.4 Subclass: <di:Image_Statistics>

This class provides various counts, metrics and conversion constants for the image contained within the primary data object.

This class contains:

- <di:bad_pixel_count>
- <di:missing_pixel_count>
- <di:despiked_pixel_count>
- <di:interpolated_pixel_count>
- <di:partial_saturated_pixel_count>
- <di:saturated_pixel_count>
- <di:adc_saturated_pixel_count>
- <di:ultracompressed_pixel_count>
- <di:scaled_maximum>
- <di:scaled_minimum>
- <di:scaled_median>
- <di:scaled_standard_deviation>
- <di:stored_maximum>
- <di:stored_minimum>
- <di:stored_median>
- <di:stored_standard_deviation>
- <di:Conversion_Constants>

<di:*_pixel_count>
The *pixel_count* attributes provide literal counts of various types of problematic pixels.

<di:scaled_*>
The *scaled* statistics are calculated by first applying scaling factors and offsets in the data to the pixel values. These differ from the PDS4 attributes found in the *<pds:Object_Statistics>* class in that overclock or reference areas that are reported as part of the data array, but are not considered part of the image itself, are excluded from the calculation of these statistics.

<di:stored_*>
The *stored* statistics are calculated from the pixel values as they are stored in the image array, *before* any scaling factor or offset is applied. These differ from the PDS4 attributes found in the *<pds:Object_Statistics>* class in that overclock or reference areas that are reported as part of the data array, but are not considered part of the image itself, are excluded from the calculation of these statistics.

<di:Conversion_Constants>

This subclass only appears in exoplanet observations from the EPOXI mission. It provides constants to be used to convert the pixel values from their stored units to other units of measure.

CHAPTER

FOUR

DEEP IMPACT/EPOXI MISSION NAMESPACE OUTLINE

<di:Mission_Parameters> is the public entry point to the Deep Impact/EPOXI Mission namespace. This class contains all other DI/EPOXI classes and must be included to gain access to them. Below is a summary outline of all classes and attributes currently available in the DI mission dictionary, in the order in which they would appear in a label if every single one was used.

Note that in real labels it is possible that not every single mission class and attribute will be found. Some attributes were not used in the primary mission, some are only applicable to exoplanet observations, etc. The point of this outline is primarily to catalog what is present and show the required ordering within classes when they are included in a label.

```
<di:Mission_Parameters>
    <di:mission_phase_name/>
    <di:spacecraft_clock_start/>
    <di:spacecraft_clock_mid/>
    <di:spacecraft_clock_stop/>

    <di:Observation_Parameters>
        <di:observation_id/>
        <di:observation_description/>
        <di:mission_activity/>
        <di:product_type/>
        <di:spacecraft_processor_id/>
        <di:sdc_pipeline_file_name/>

        <di:Imaging>
            <di:image_number/>
            <di:commanded_image_count/>
            <di:image_compression_history/>

        <di:Geometry>
            <di:geometry_quality/>
            <di:Spacecraft_Rotation_Velocity_Vector>
                <di:x_axis_rotation unit="rad/s"/>
                <di:y_axis_rotation unit="rad/s"/>
                <di:z_axis_rotation unit="rad/s"/>

    <di:Instrument_Parameters>
        <di:instrument_mode/>
        <di:instrument_mode_mnemonic/>
        <di:instrument_mode_id/>

        <di:Instrument_Temperatures>
```

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```
            <di:Temperature_Measurement>
                <di:sensor_location/>
                <di:temperature unit="K"/>
            <di:Temperature_Measurement>
            ...

        <di:Instrument_Voltages>
            <di:Voltage_Measurement>
                <di:sensor_location/>
                <di:voltage unit="V"/>
            <di:Voltage_Measurement>
            ...

    <di:Image_Statistics>
        <di:bad_pixel_count/>
        <di:missing_pixel_count/>
        <di:despiked_pixel_count/>
        <di:interpolated_pixel_count/>
        <di:partial_saturated_pixel_count/>
        <di:saturated_pixel_count/>
        <di:adc_saturated_pixel_count/>
        <di:ultracompressed_pixel_count/>

        <di:scaled_maximum/>
        <di:scaled_minimum/>
        <di:scaled_median/>
        <di:scaled_standard_deviation/>

        <di:stored_maximum/>
        <di:stored_minimum/>
        <di:stored_median/>
        <di:stored_standard_deviation/>

        <di:Conversion_Constants>
            <di:data_to_calibrated_dn_multiplier/>
            <di:data_to_ioverf_multiplier/>
            <di:data_to_radiance_multiplier/>
```

CHAPTER

FIVE

ALPHABETICAL LIST OF CLASSES

A complete list of all classes in the Deep Impact/EPOXI Mission Dictionary, in alphabetical order, is available through the PDS4 Data Dictionary page, which is regenerated automatically with each release of the PDS4 Information Model.

To find the Deep Impact/EPOXI Mission class list, look down the list of (alphabetically sorted) dictionary prefixes in the left menu for "Classes in the di namespace". Select that item and the list of classes will be presented on both the left and the right as clickable links.

Clicking on the specific class name will produce a grid with the full, formal definition of the class.

Clicking on the class name in the "Referenced from:" line at the bottom of the grid will take you to the containing class, where you can see the cardinality of the class (i.e., whether it is required, optional, or repeatable) in the containing class.

You can also click on the attribute names listed to see details of the attribute definitions.

CHAPTER

SIX

ALPHABETICAL LIST OF ATTRIBUTES

A complete list of all attributes in the Deep Impact/EPOXI Mission Dictionary, in alphabetical order, is available through the PDS4 Data Dictionary page, which is regenerated automatically with each release of the PDS4 Information Model.

To find the Deep Impact/EPOXI Mission attribute list, look down the list of (alphabetically sorted) dictionary prefixes in the left menu for "Attributes in the di namespace". Select that item and the list of attributes will be presented on both the left and the right as clickable links.

Clicking on the specific attribute name will produce a grid with the full, formal definition of the attribute, including data type, restrictions on values, and the list of defined permissible values (if any) and their definitions.

Note that attributes might appear as members of different classes, and that their definitions, or more likely their permissible values, might be context-dependent.

Clicking on the class name in the title bar of the attribute grid will take you to the definition of the class containing that attribute.

If the attribute has an associated unit of measure type, that attribute *must* have an XML attribute called "unit" in its tag when it is used. For example:

```
<di:temperature unit="K">40.4694</di:temperature>
```

You can see valid values to use for the "unit=" XML attribute by clicking on the value of "Unit of Measure Type" in the grid.

CHAPTER

SEVEN

MOCKUP LABELS

Mockup labels are created during the initial design and test phase for the conversion from PDS3 to PDS4. They are not particularly good templates for designing a *real* label, in that they usually omit the half dozen or so discipline dictionaries needed to provide complete metadata for an actual observational product. But they do contain plausible values for the specific classes illustrated, as found in a sample product from the mission.

Mockup labels can be found in the Deep Impact/EPOXI Mission Dictionary GitHub repo: https://github.com/pds-data-dictionaries/ldd-di. They are in the *test/examples/* directory. Available mockups are:

- First draft label for general design (Design1)